Figure 28: A tied package can be split across multiple counterparties, with each counterparty receiving its corresponding fraction of the option trade and hedge trade.

15.10 Why Delta Matters So Much

Delta is the first Greek because it is the clearest bridge between pricing and trading.

It tells us:

- how an option reacts to a small underlying move,
- how to hedge the option,
- how directional the current book is,
- and roughly how likely an option is to finish in the money.

Remark 15.11. In practice, market makers constantly track both the delta of individual options and the net delta of the full portfolio. The portfolio view is what determines how much underlying they need to buy or sell to stay balanced.

16 Gamma

16.1 Definition and Core Intuition

Definition 16.1 (Gamma)**.** **Gamma**, denoted by Γ, is the rate of change of delta with respect to changes in the underlying price.

Equivalently, gamma tells us how much an option's delta will change for a 1-point move in the underlying.

Remark 16.1. If delta is the first directional sensitivity of an option, then gamma measures how unstable or fast-changing that directional sensitivity is.

Example 16.1 (Basic gamma interpretation)**.** Suppose an option has delta 0.30 and gamma 0.02.

If the underlying rises by 1 point, the option's delta increases approximately to

$$0.30 + 0.02 = 0.32.$$

If the underlying falls by 1 point, the option's delta decreases approximately to

$$0.30 - 0.02 = 0.28.$$

16.2 Why Gamma Matters

Gamma matters because delta is not constant. As the underlying moves, the option's directional exposure changes. A market maker who is delta-neutral at one price will generally stop being delta-neutral after the underlying moves.

Remark 16.2. This is why gamma is so operationally important. Delta tells us our current exposure. Gamma tells us how fast that exposure will change once the market starts moving.

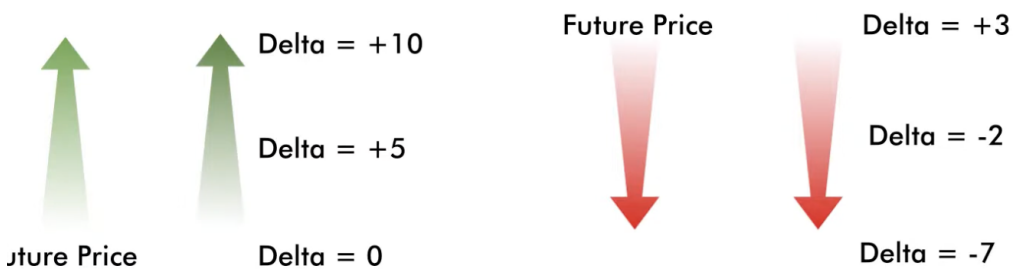

Figure 29: Gamma changes delta in the same direction as the underlying move. When the underlying rises, delta becomes more positive; when the underlying falls, delta becomes more negative.

16.3 Long Gamma and Short Gamma

All outright options have positive gamma. Therefore:

- if you are **long** an option, you are long gamma,
- if you are **short** an option, you are short gamma.

A long gamma position gains deltas in the direction of the underlying move:

- if the underlying rises, you get longer delta,
- if the underlying falls, you get shorter delta.

A short gamma position experiences the exact opposite.

Remark 16.3. This is the core asymmetry of gamma. Long gamma helps you adapt with the move. Short gamma fights the move.

16.4 Gamma as the Curvature of Option Value

Gamma is also what creates curvature in option prices. Delta gives a local linear approximation, but gamma explains why that linear approximation changes as the underlying moves.

Example 16.2 (Price change with changing delta)**.** Suppose the future starts at 30.00, the option price is 0.91, delta is 0.31, and gamma is 0.06.

If the future rises to 31.00, the delta rises from 0.31 to about 0.37. A better approximation to the option price move is to use the average delta across the move:

$$\frac{0.31 + 0.37}{2} = 0.34.$$

So the option price increases by about 0.34, giving a new price near

$$0.91 + 0.34 = 1.25.$$

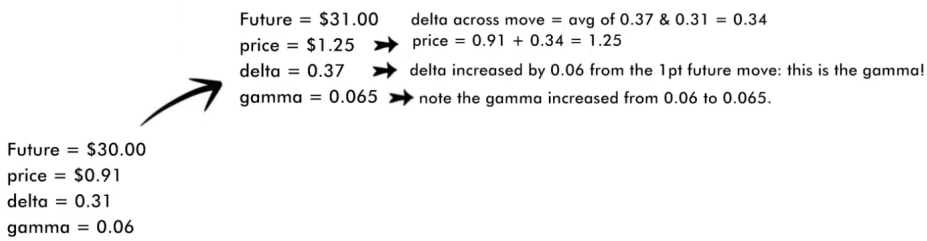

Figure 30: Example of gamma changing delta during an underlying move. The option's price change is better approximated using the average delta across the move rather than the starting delta alone.

Remark 16.4. This is why delta-only approximations break down for larger moves. Gamma makes delta itself move during the path.

16.5 Gamma and Re-Hedging

Because gamma changes delta, an option portfolio must be re-hedged after the underlying moves.

Definition 16.2 (Re-hedging)**. Re-hedging** means trading the underlying again after a move in order to bring the portfolio back toward delta-neutral.

Example 16.3 (Gamma re-hedging intuition)**.** Suppose a trader is long options with total option delta +50, and has sold 50 futures to create a net delta of 0.

If the underlying rises and the option position picks up +5 more deltas from gamma, then the portfolio is now net +5 delta. To get back to neutral, the trader sells 5 more futures.

16.6 Gamma Scalping

This repeated process of re-hedging after moves is what traders call **gamma scalping**.

Definition 16.3 (Gamma scalping)**. Gamma scalping** is the process of repeatedly trading the underlying against a long-gamma options position as delta changes through market movement.

The standard long-gamma pattern is:

- market goes up → you become longer delta → you sell futures,
- market comes back down → you become shorter delta → you buy futures back.

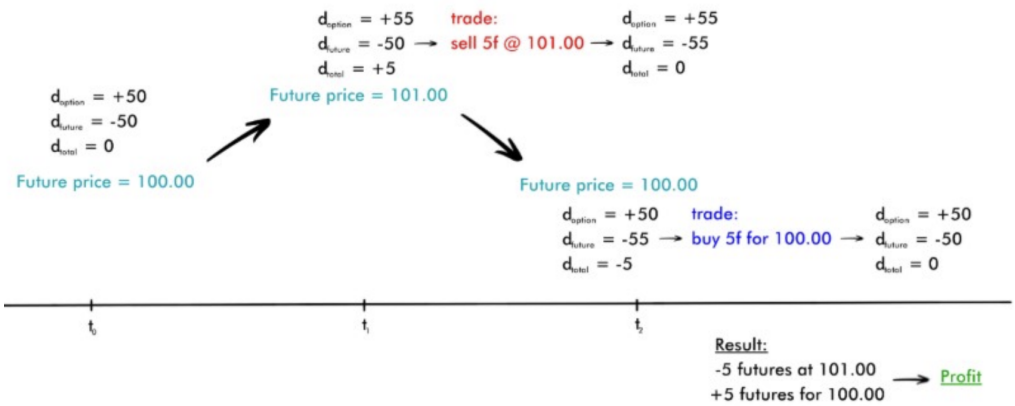

Figure 31: Gamma re-hedging example. A long-gamma trader who picks up extra delta after an upward move can re-hedge by selling futures, and later buy them back if the market reverses.

So if the market mean-reverts, the long-gamma trader tends to:

sell high, buy low.

That creates trading profit in the hedge.

Remark 16.5. Gamma scalping is not magic. The hedge profit must still be compared against theta decay. A long-gamma position can scalp well and still lose overall if too much option value decays through time.

16.7 Where Gamma Is Largest

Gamma is generally largest for options that are near the money, especially when expiration is close.

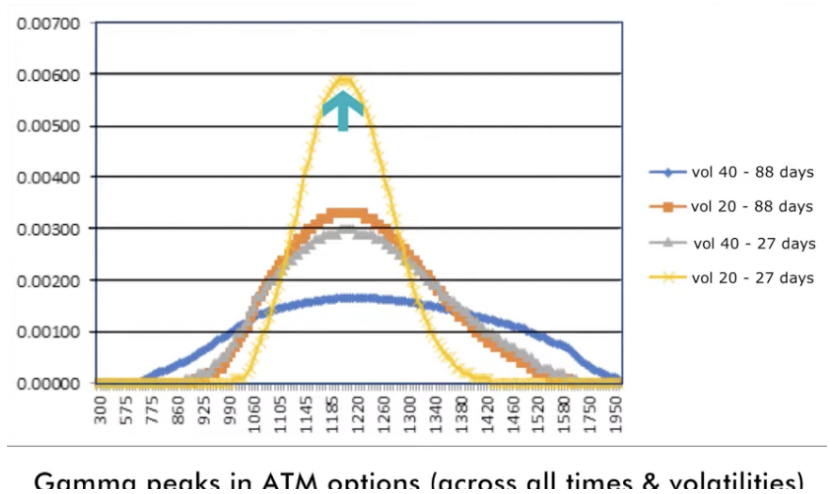

Figure 32: Gamma peaks near at-the-money options across different times to expiry and volatility assumptions.

Remark 16.6. Near expiration, the option's final state becomes more binary. That makes delta change very rapidly around the at-the-money region, which is exactly why gamma spikes there.

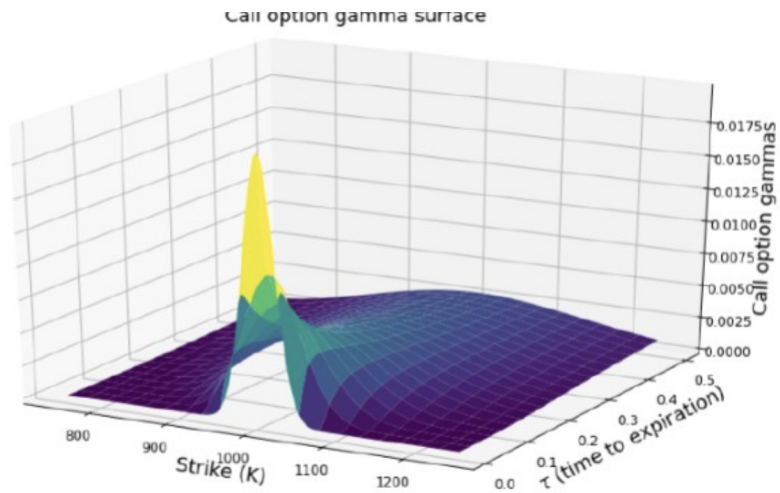

Figure 33: Gamma surface across strike and time to expiry. Gamma is most concentrated near at-the-money strikes and near expiration.

16.8 Effect of Time and Volatility on Gamma

Gamma does not behave uniformly across all strikes.

Two broad principles are most useful:

- gamma is typically largest near the money,
- gamma generally becomes more concentrated near expiry.

Lower-volatility and shorter-dated options tend to have sharper gamma concentration near the at-the-money region. Intuitively, when expiration is close, a small move in the underlying can change the probability of finishing in the money very quickly, so delta must adjust more aggressively.

Remark 16.7. Gamma is not flat across strike space. It is highest where the option is most sensitive to whether it finishes in or out of the money, which is why at-the-money options dominate most gamma discussions.

16.9 Gamma and the Shape of the Surface

Looking across both strike and maturity, gamma forms a surface rather than a single curve. Near-the-money options with little time remaining tend to sit at the highest part of that surface.

A second way to view the same idea is to compare gamma curves under different volatility and time assumptions. The central pattern remains the same: gamma peaks around ATM options.

16.10 Long Gamma and Portfolio Management

The practical importance of gamma appears once we hedge a delta position. A trader may begin delta-neutral, but gamma guarantees that this neutrality will not survive a move in the underlying.

Suppose we own a portfolio of options with total delta +50. To neutralize that directional exposure, we sell 50 futures. The net portfolio delta is then approximately zero.

If the underlying rises and our option position picks up +5 more deltas through gamma, then the portfolio is no longer flat. It is now net +5 deltas, so we must sell 5 more futures to restore neutrality.

If the underlying later falls back and those deltas disappear, we must buy those futures back. In that case we have sold futures at a higher price and bought them back at a lower price.

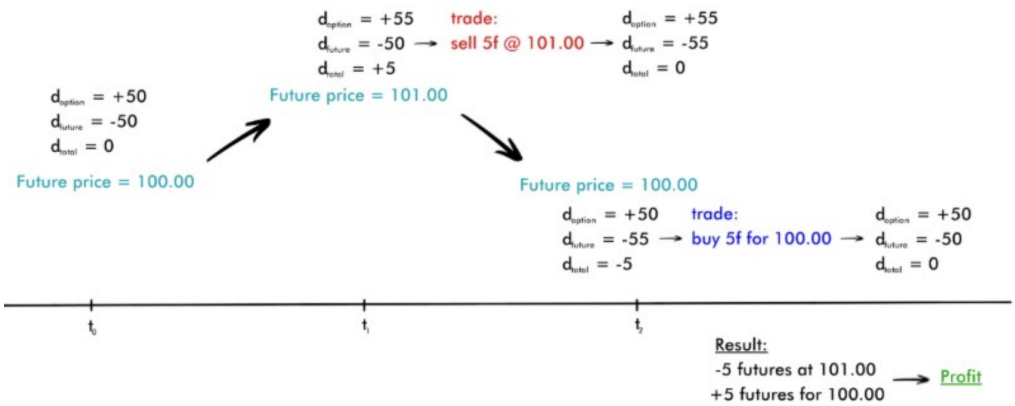

Figure 34: Gamma re-hedging example. A long-gamma trader gains delta in the direction of the move, then re-hedges by trading futures.

Definition 16.4 (Gamma Scalping)**. Gamma scalping** is the repeated re-hedging of a long-gamma or short-gamma portfolio as the underlying moves and delta changes.

For a **long gamma** trader, this usually means:

- selling futures after the market rises,
- buying futures after the market falls.

That is the favorable pattern:

$$\text{sell high, buy low.}$$

For a **short gamma** trader, the pattern reverses:

- buying futures after the market rises,
- selling futures after the market falls.

That is the unfavorable pattern:

$$\text{buy high, sell low.}$$

Remark 16.8. This is the real trading intuition behind gamma. Long gamma helps the hedge. Short gamma hurts the hedge.

16.11 A Small Numerical Example

Assume a call has:

$$\text{price} = 0.91, \qquad \Delta = 0.31, \qquad \Gamma = 0.06,$$

with the future initially at 30.00.

If the future rises to 31.00, then the option's delta rises approximately from

$$0.31 \to 0.37.$$

A better approximation to the option price change uses the average delta over the move:

$$\frac{0.31 + 0.37}{2} = 0.34.$$

So the option price moves by about

$$1 \times 0.34 = 0.34,$$

which gives a new option price near

$$0.91 + 0.34 = 1.25.$$

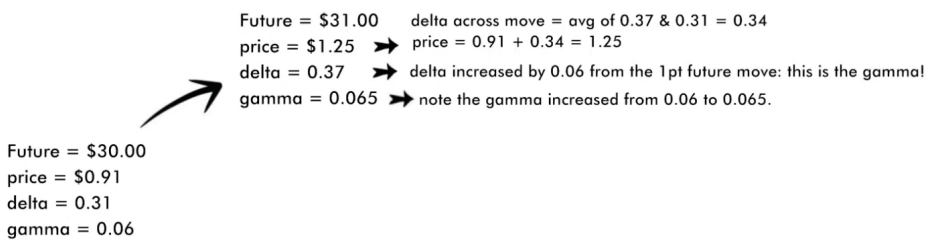

Figure 35: Example of gamma changing delta during an underlying move. The option price change is better approximated with the average delta across the move.

Remark 16.9. Delta alone gives a linear approximation. Gamma explains why that approximation bends as the underlying moves.

16.12 Cash Gamma

In practice, traders often normalize gamma into a dollar-sized risk number called **cash gamma**.

Definition 16.5 (Cash Gamma)**.** **Cash gamma** is the amount by which the portfolio's cash delta changes for a 1% move in the underlying.

This is useful because raw gamma in "deltas per point" is not directly comparable across products with very different prices or multipliers.

If one future has cash delta

$$\text{Cash Delta} = (\text{future price}) \times (\text{multiplier}) \times 1,$$

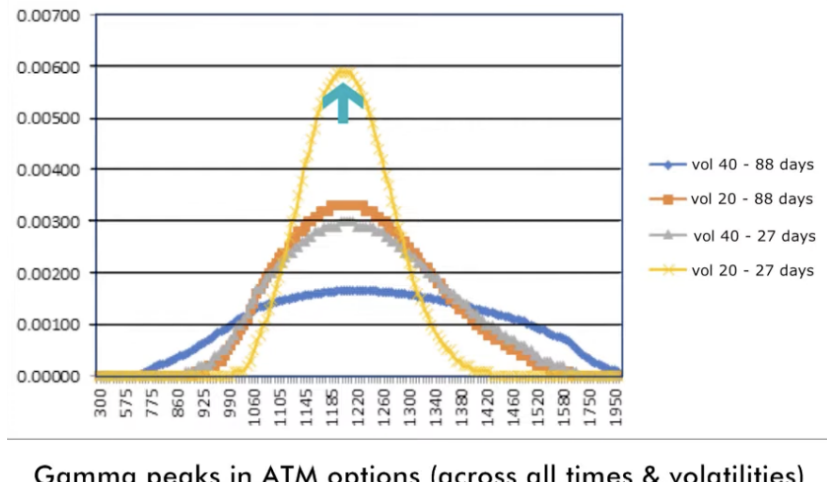

Figure 36: Gamma peaks near at-the-money options across different volatility and maturity assumptions.

then cash gamma for a 1% move can be interpreted as:

$$\text{Cash Gamma} = (\text{deltas picked up on a 1\% move}) \times (\text{cash delta of one future}).$$

Example 16.4 (Cash gamma intuition)**.** Suppose a future is trading at 50, the contract multiplier is 1000, and a position picks up 3 deltas for a 1% move.

Then one future has cash delta

$$50 \times 1000 = 50{,}000.$$

So the cash gamma is

$$3 \times 50{,}000 = 150{,}000.$$

Remark 16.10. Cash gamma is a sizing tool. It tells the trader not just that gamma exists, but how large the re-hedging pressure will be in dollars.

16.13 Why Traders Care So Much About Gamma

Gamma is the Greek that explains why option books do not stay still.

It tells us:

- how quickly delta will change,
- how often the portfolio will need re-hedging,
- whether the trader benefits or suffers from large realized moves in the underlying.

This is also why gamma is inseparable from hedging. A trader does not manage gamma in the abstract. Gamma shows up as repeated hedge adjustments in the underlying.

16.14 Gamma Versus Theta

At first glance, long gamma sounds ideal. A long-gamma trader gains delta in the direction of the move and can profit from re-hedging. So why not always stay long gamma?